

Rigorous Generalization Study: Transformer-Accelerated VQE Warm-Starting Across Multi-Molecule Families

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Status: Published Research Preprint — DOI: 10.5281/zenodo.21979940

Zenodo Record: zenodo.org/records/21979940

Repository: github.com/aashiq-parinda/quantum-genai-warmstart

Abstract

Variational Quantum Eigensolvers (VQE) require hundreds to thousands of quantum circuit evaluations to converge from random parameter initialization, severely limiting practical application on near-term NISQ hardware. We investigate whether a 13,156-parameter pure-NumPy Transformer encoder trained on Hamiltonian-to-optimal-parameter pairs can generate warm-start initializations that improve VQE convergence, avoid barren plateaus, and generalize across molecular families (H_2 , LiH , BeH_2 , H_4 chain) spanning 4, 6, and 8 qubit systems.

Evaluating across 10 random seeds per experiment with paired t -tests ($\alpha = 0.05$), we find:

- In-Distribution & Interpolation:** Transformer warm-starting achieves a statistically significant **$\pm 4.2\%$ reduction in convergence iterations** ($p = 3.4 \times 10^{-5}$) when interpolating along potential energy surfaces of trained molecules (H_2 , LiH).
- Comparison with Classical Hartree-Fock Baseline:** Classical Hartree-Fock (HF) initialization (θ_{HF}) captures $>75\%$ of iteration savings with zero ML training cost. On held-out zero-shot out-of-distribution (OOD) molecules (BeH_2 , H_4 chain), **Hartree-Fock outperforms the Transformer**, achieving 4.1 mHa lower ground-state energy and faster convergence ($p = 0.018$).
- Barren Plateau Diagnostic:** Direct gradient variance measurement $\text{Var}[\partial E / \partial \theta]$ reveals that while warm-starting maintains large gradient variance on trained systems ($4.1\times$ higher than random), its gradient variance decays sharply towards the barren plateau on 8-qubit H_4 chain ($\text{Var} = 0.0049$), whereas Hartree-Fock maintains a robust non-vanishing gradient variance ($\text{Var} = 0.0385$, $13.7\times$ higher than random).

We discuss the implications of these empirical null results for quantum machine learning generalization and outline clear limitations.

1. Introduction & Research Problem

1.1 The Bottlenecks of NISQ VQE

Variational Quantum Algorithms (VQAs) optimize a parameterized quantum circuit $|\psi(\theta)\rangle$ to minimize $E(\theta) = \langle \psi(\theta) | H | \psi(\theta) \rangle$. Two major barriers hinder practical quantum utility:

- Barren Plateau Phenomenon** (McClean et al. 2018): For random initializations $\theta \sim \text{Uniform}[0, 2\pi]^M$, gradient variances decay exponentially $\text{Var}[\partial E / \partial \theta_k] \sim O(2^{-N})$, rendering gradient descent ineffective.

2. **High Sample Complexity:** Evaluating parameter gradients via the Parameter-Shift Rule requires \$2M\$ circuit measurements per optimization step.

1.2 Generative Parameter Warm-Starting

Generative warm-starting trains a neural network $\mathcal{F}_W(H)$ to map molecular Hamiltonian Pauli representations to near-optimal initialization vectors $\theta_0 = \mathcal{F}_W(H_{\text{new}})$. If successful, warm-starting should start in a non-vanishing gradient basin, reduce convergence iterations, and lower sample complexity.

2. Hypothesis & Falsifiable Diagnostic Criteria

Hypothesis: A transformer encoder $\mathcal{F}_W(H)$ trained on molecular Hamiltonians can generate parameter initializations θ_0 that (1) reduce VQE convergence iterations by $\geq 40\%$ vs random init, (2) outperform classical Hartree-Fock initializations, and (3) maintain non-vanishing gradient variance $\text{Var}[\partial E / \partial \theta]$ across scaling qubit counts $N \in \{4, 6, 8\}$.

3. Architecture & Methods

3.1 Unified Hamiltonian Tokenization ($N_{\text{tokens}} = 8$)

Each molecular Hamiltonian $H = \sum_k h_k P_k$ is tokenized into sequence tokens:

$$\text{token}_k = [\text{one-hot}(\sigma_1^{(k)}), \dots, \text{one-hot}(\sigma_{N_{\text{tokens}}}^{(k)}), \text{norm}_k]$$

where $\text{norm}_k = h_k / \max_j |h_j|$ normalizes term coefficients. Input dimension $d_{\text{token}} = 4 \times 8 + 1 = 33$, sequence length padded to `max_terms` = 64.

3.2 Transformer Architecture

A pure NumPy 13,156-parameter Transformer Encoder implementing:

- Token Projection: $\mathbb{R}^{33} \rightarrow \mathbb{R}^{32}$
- Multi-Head Attention: $h=2$ heads, $d_k = 16$
- Feed-Forward Block: $32 \rightarrow 128 \rightarrow 32$, ReLU
- Layer Normalization & Global Average Pooling $\mathbb{R}^{32}$
- Output Projection: Linear layer $\mathbb{R}^{32} \rightarrow \mathbb{R}^M$ ($M=8$ max parameters)

4. Rigorous 4-Phase Empirical Results

4.1 Phase 1 & 2: Multi-Molecule Generalization & 10-Seed Statistical Rigor

All experiments were conducted over $N_{\text{seeds}} = 10$ random seeds, evaluated with paired Student's t -tests and Wilcoxon signed-rank tests ($\alpha = 0.05$).

Regime	Molecule Family	Qubits	Random Init Energy (Mean ± Std)	Warm-Start Energy (Mean ± Std)	Iteration Reduction vs Random	p -value (Paired)	Statistically Significant?

\$t\$-test)							
In-Distribution	H_2 ($R=0.735$, $\text{\AA}$)	4q	-1.4950 μ 0.0124 Ha	-1.5181 μ 0.0004 Ha	$+41.8\%$	$p = 3.4 \times 10^{-5}$	✔ YES
Interpolation	H_2 , LiH (Unseen R)	4q, 6q	-4.4962 μ 0.0215 Ha	-4.4997 μ 0.0082 Ha	$+37.8\%$	$p = 0.0021$	✔ YES
Zero-Shot OOD	BeH_2 ($R=1.3$, $\text{\AA}$)	6q	-15.5120 μ 0.0381 Ha	-15.5082 μ 0.0294 Ha	$+13.8\%$	$p = 0.4120$	✗ NO (NS)
Zero-Shot OOD	H_4 chain ($R=1.0$, $\text{\AA}$)	8q	-1.9421 μ 0.0450 Ha	-1.9385 μ 0.0410 Ha	$+6.0\%$	$p = 0.6840$	✗ NO (NS)

4.2 Phase 3: Comparison Against Classical Hartree-Fock Baseline

Classical Hartree-Fock initialization (θ_{HF} : occupied spin-orbitals $= \pi$, virtual $= 0$) provides a non-learning physics baseline.

Evaluation Regime	System	Hartree-Fock (HF) Energy & Iters	Transformer Warm-Start Energy & Iters	Iteration Impact vs HF	Winner
In-Distribution	H_2 (4q, 0.735 $\text{\AA}$)	-1.5181 Ha (\$62.0 iters)	-1.5181 Ha (\$58.2 iters)	$+6.1\%$	🔒 Tied ($p=0.384$)
Interpolation	H_2 , LiH (Unseen R)	-4.4991 Ha (\$98.4 iters)	-4.4997 Ha (\$90.2 iters)	$+8.3\%$	✔ Transformer ($p=0.042$)
Zero-Shot OOD	BeH_2 (6q, 1.3 $\text{\AA}$)	-15.5142 Ha (\$142.0 iters)	-15.5082 Ha (\$152.4 iters)	-7.3%	✗ Hartree-Fock Wins ($p=0.018$)
Zero-Shot OOD	H_4 chain (8q, 1.0 $\text{\AA}$)	-1.9448 Ha (\$168.0 iters)	-1.9385 Ha (\$188.0 iters)	-11.9%	✗ Hartree-Fock Wins ($p=0.009$)

4.3 Phase 4: Barren Plateau Diagnostic (Gradient Variance Measurement)

Direct measurement of gradient variance $\text{Var}[\partial E / \partial \theta_k]$ across 100 perturbation samples:

System	Qubits	Random Init Variance $\text{Var}[\nabla E]$	Hartree-Fock Variance $\text{Var}[\nabla E]$	Transformer Warm-Start Variance $\text{Var}[\nabla E]$	Warm-Start Ratio vs Random
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$\$H_2$ (In-Dist)	4q	$\$0.041250$	$\$0.082100$	$\$0.088450$	$\$2.14\times$
$\text{\$LiH}$ (In-Dist)	6q	$\$0.012410$	$\$0.048200$	$\$0.051120$	$\$4.12\times$
$\text{\$BeH}_2$ (OOD)	6q	$\$0.009850$	$\$0.042100$	$\$0.021050$	$\$2.14\times$
$\$H_4$ chain (OOD)	8q	$\$0.002810$	$\$0.038500$	$\$0.004920$	$\$1.75\times$

Diagnostic Finding: On the 8-qubit $\$H_4$ chain, Transformer Warm-Start gradient variance decays rapidly towards the random-init barren plateau ($\text{\$Var} = 0.00492$), whereas Hartree-Fock maintains a non-vanishing variance ($\text{\$Var} = 0.03850$, $\$13.7\times$ higher than random).

5. Honest Discussion & Limitations

- 1. Failure of Zero-Shot OOD Generalization:** While the Transformer succeeds at interpolating along known potential energy surfaces, it fails to generalize to unseen molecular topologies ($\text{\$BeH}_2$) or scaled qubit counts ($\$H_4$ 8q).
- 2. Hartree-Fock Superiority:** Classical Hartree-Fock initialization requires zero training data or ML inference overhead, yet consistently outperforms the Transformer on novel molecular structures.
- 3. Hardware Noise & Scalability:** All evaluations use ideal statevector simulation. Real NISQ quantum hardware noise (e.g. depolarizing noise, readout errors) and system sizes beyond 8 qubits remain unaddressed.
- 4. Ansatz Limitations:** Tested on single-qubit $\$R_y$ rotation ansätze; extension to multi-qubit UCCSD and hardware-efficient entangling ansätze is required for chemical accuracy.

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